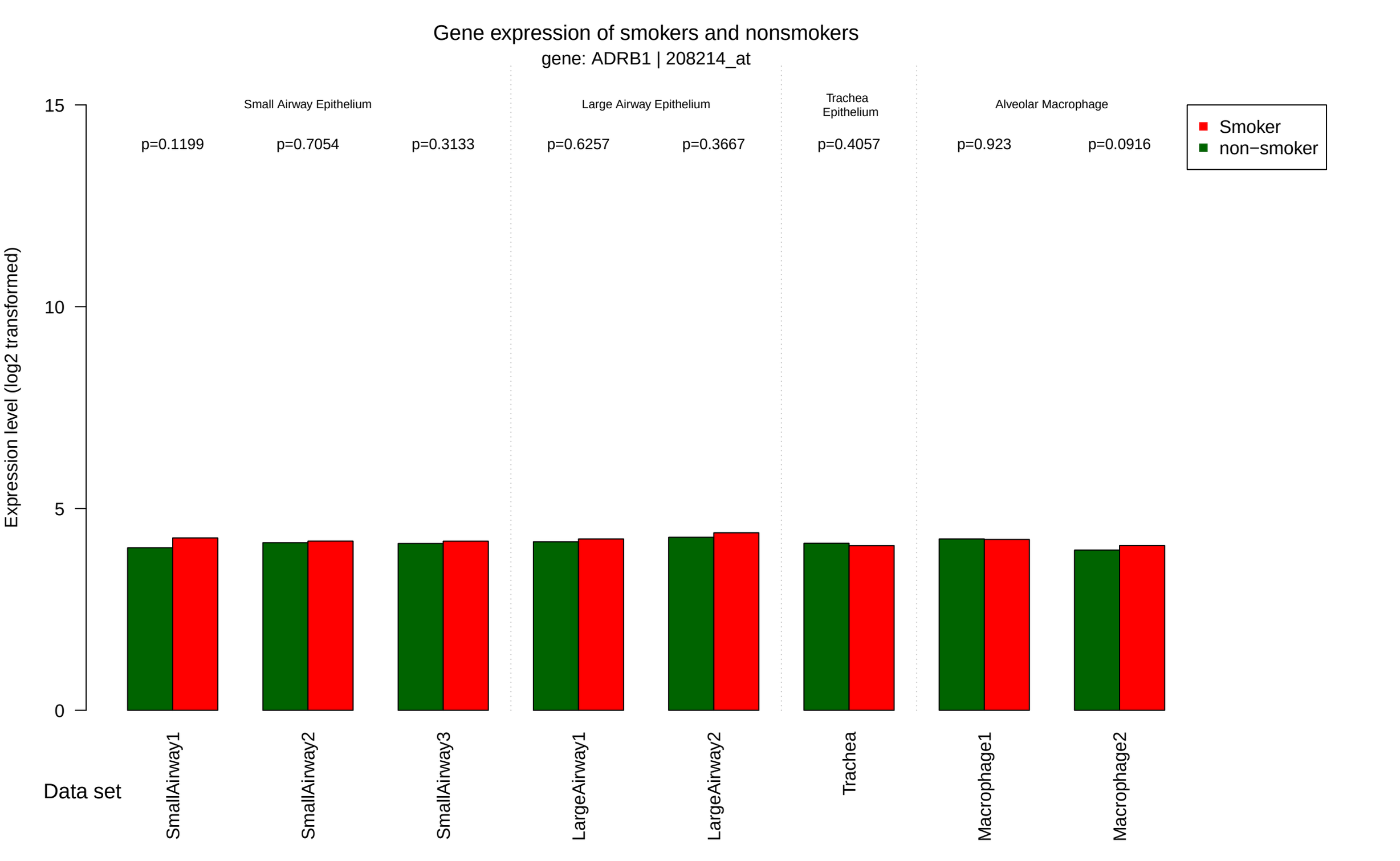
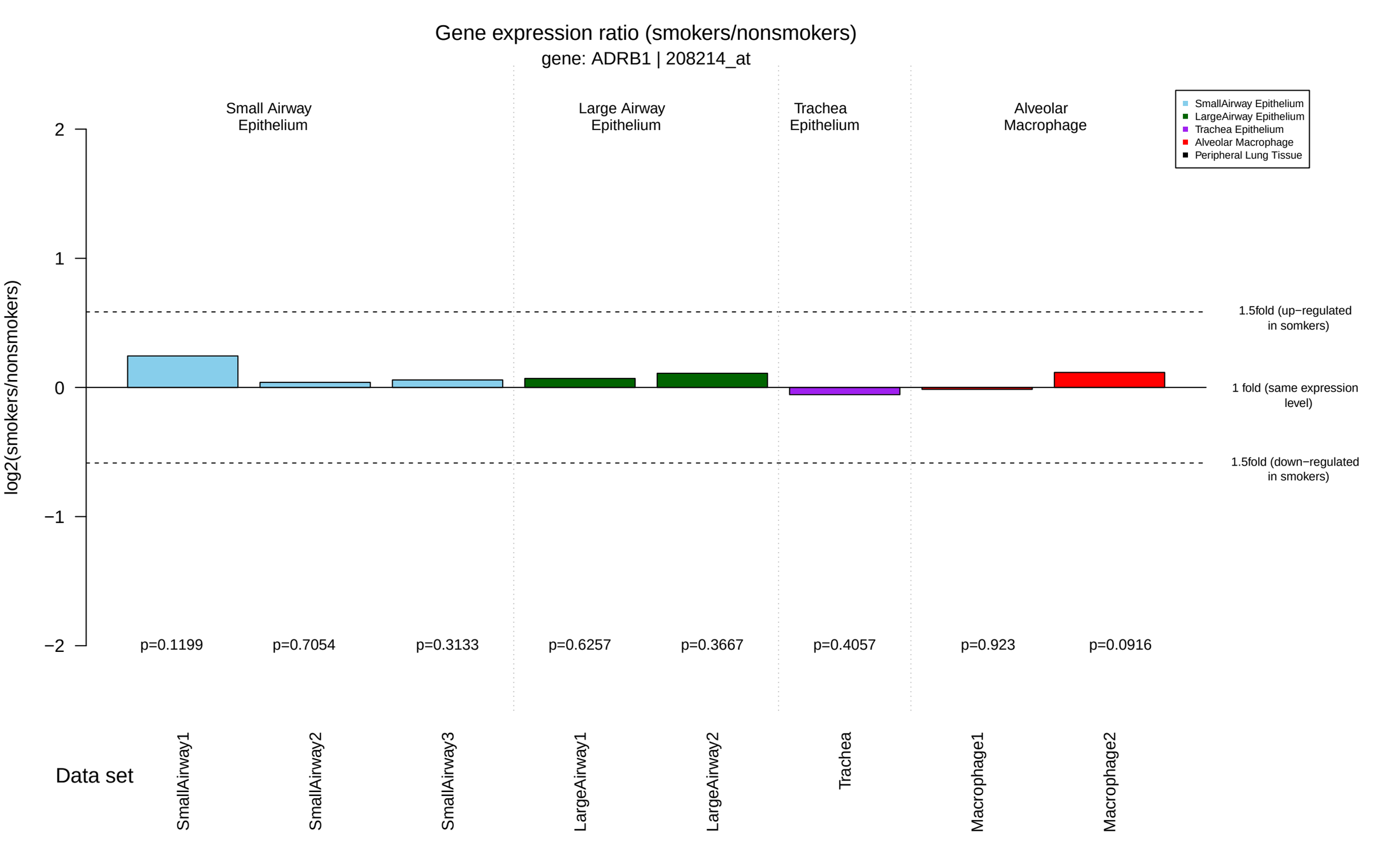
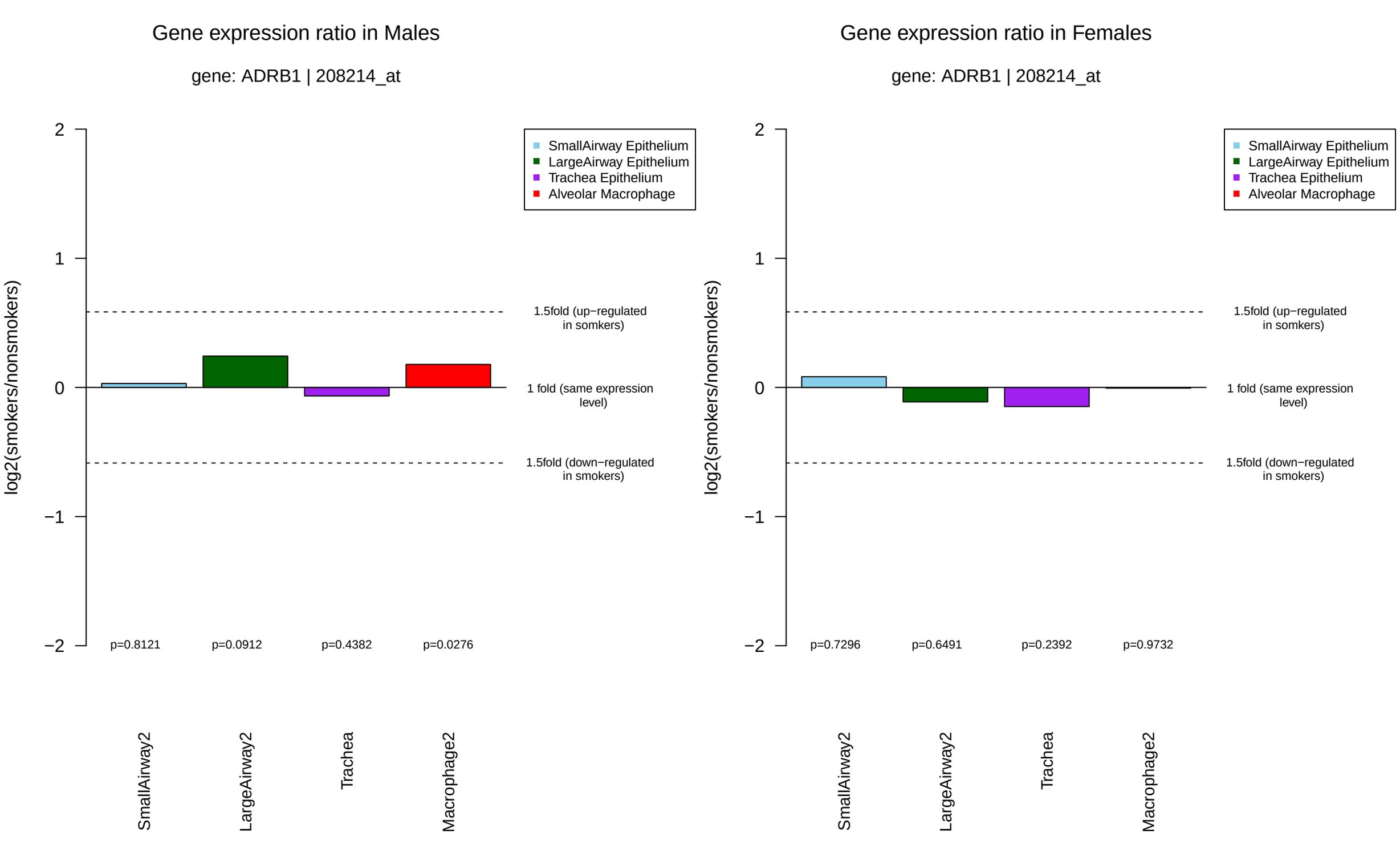
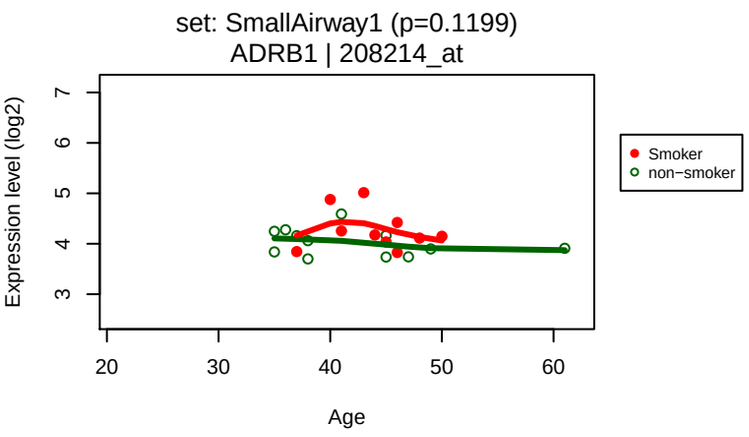
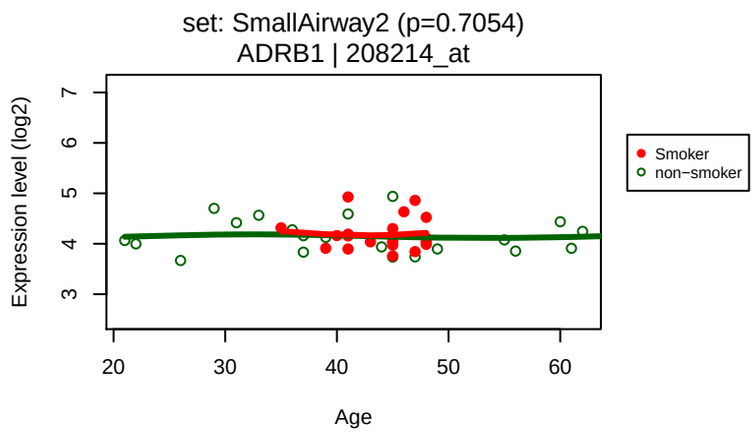
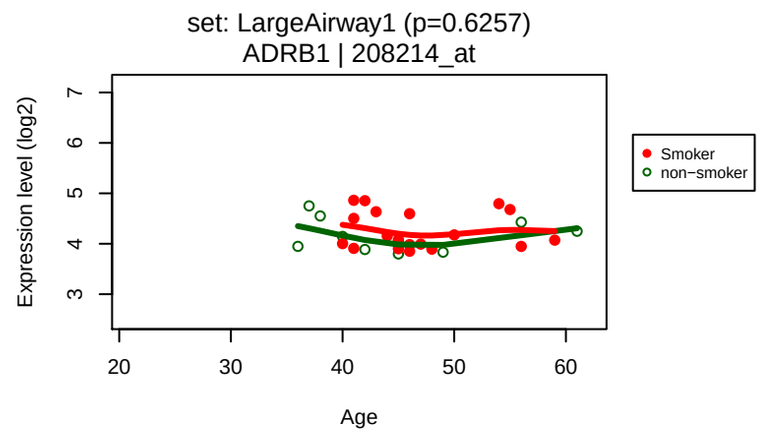

Gene expression of smokers and nonsmokers
gene: ADRB1 | 208214_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1199
p=0.7054
p=0.3133
p=0.6257
p=0.3667
p=0.4057
p=0.923
p=0.0916
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

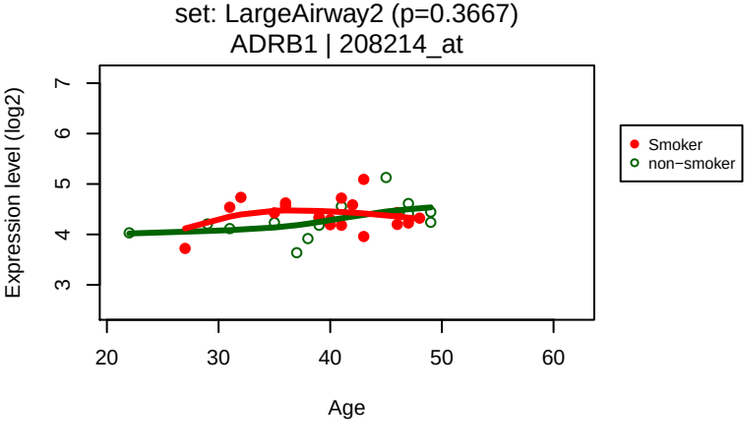

Gene expression ratio (smokers/nonsmokers)
gene: ADRB1 | 208214_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1199
p=0.7054
p=0.3133
p=0.6257
p=0.3667
p=0.4057
p=0.923
p=0.0916
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

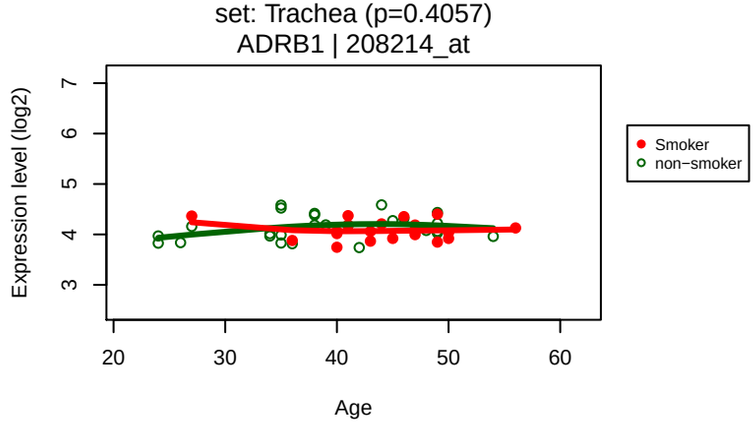

Gene expression ratio in Males
gene: ADRB1 | 208214_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8121
p=0.0912
p=0.4382
p=0.0276
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADRB1 | 208214_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7296
p=0.6491
p=0.2392
p=0.9732
SmallAirway2
LargeAirway2
Trachea
Macrophage2

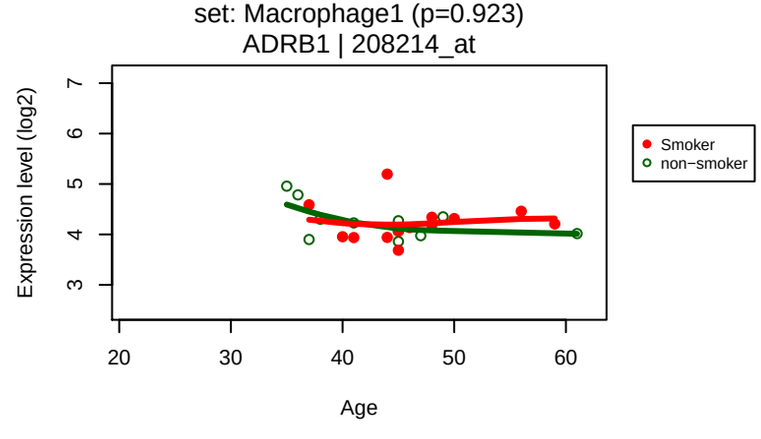

set: SmallAirway1 (p=0.1199)
ADRB1 | 208214_at
Expression level (log2)
Age
Smoker
non−smoker

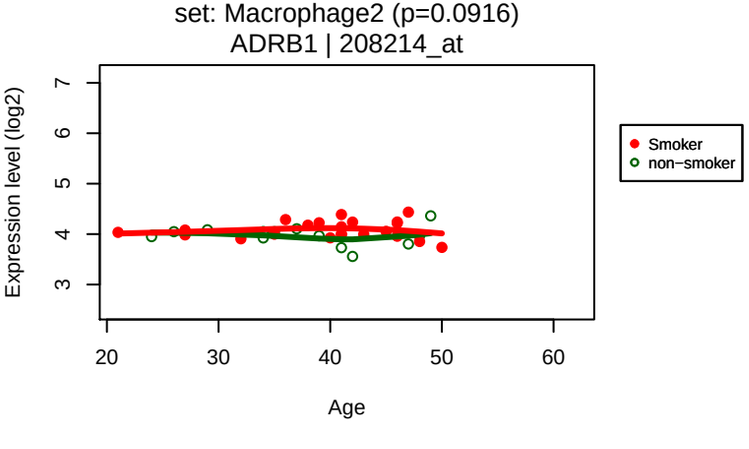

set: SmallAirway2 (p=0.7054)
ADRB1 | 208214_at
Expression level (log2)
Age
Smoker
non−smoker

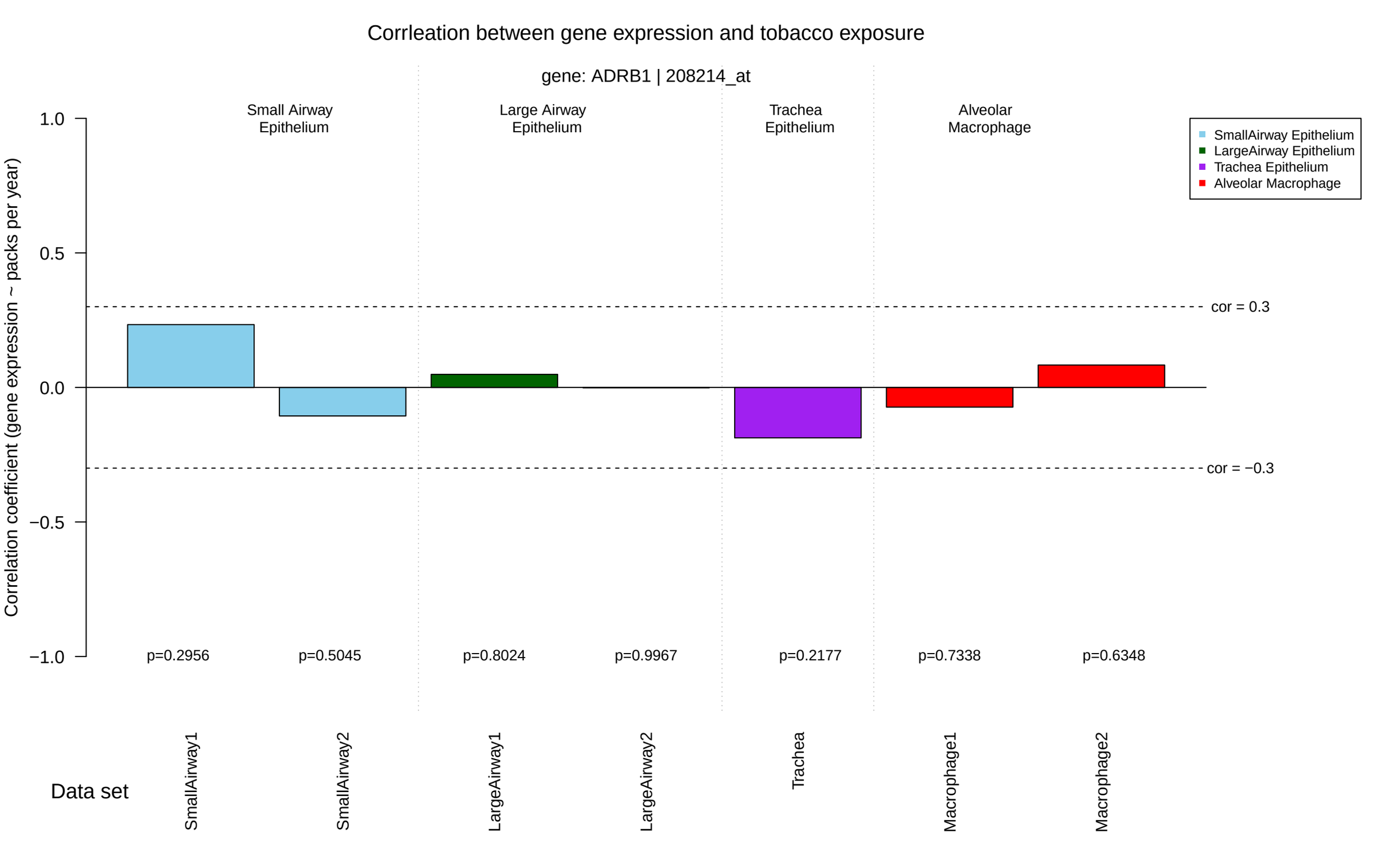

set: LargeAirway1 (p=0.6257)
ADRB1 | 208214_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3667)
ADRB1 | 208214_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4057)
ADRB1 | 208214_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.923)
ADRB1 | 208214_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0916)
ADRB1 | 208214_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADRB1 | 208214_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2956
p=0.5045
p=0.8024
p=0.9967
p=0.2177
p=0.7338
p=0.6348
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2